

PCAP-Lab-2 Submission

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Q1:

```
● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ gcc -fopenmp q1.c -o q1
● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ ./q1
Enter size of the matrix:
3 3
Enter matrix A:
1 2 3 4 5 6 7 8 9
Enter matrix B:
9 8 7 6 5 4 3 2 1
Resultant:
10      10      10
10      10      10
10      10      10
Serial execution time = 0.000001
Parallel execution time = 0.005337
```

Q2:

```
● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ gcc -fopenmp q2.c -o q2
● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ ./q2
Enter 5x5 matrix A:
1 2 3 4 5
6 7 8 9 0
23 34 23 22 6
8 63 923 7 3
67 92 45 6 1
Row-0 => Thread-0
Row-2 => Thread-2
Row-3 => Thread-3
Row-1 => Thread-1
Row-4 => Thread-4

Matrix B:
0      1      1      1      1
9      0      0      0      0
34     34     0      6      6
923    923    923    0      3
92     92     92     92     0
```

Q3:

```

● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ gcc q3.c -fopenmp -lm -o q3
● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ ./q3
Enter matrix dimensions:
3 3
Enter Matrix A:
1 2 3 4 5 6 7 8 9

Matrix B:
1      2      3
4      10     6
7      8      9

Matrix D:
1      2      3
4      2      6
7      8      9

```

Q4:

```

● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ gcc q4.c -fopenmp -o q4
● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ ./q4
Enter matrix size: 3
Enter matrix:
1 2 3 4 5 6 7 8 9
Enter vector:
5 6 4
Result:
29
74
119
Execution time = 0.006416 seconds

```

Q5:

```

● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ gcc q5.c -fopenmp -o q5
● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ ./q5
Enter array size: 3
Array generated by Thread-14
Element 1 processed by Thread-1
Element 2 processed by Thread-2
Element 0 processed by Thread-0

Original      Square
3             9
6             36
7             49

```

Q6: (Including performance analysis)

```
● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ gcc q6.c -fopenmp -o q6
● 5aimlb1@db1-06:~/Documents/PCAP_Dhanunjay_240962018/Lab2$ ./q6
Enter vector size: 3
Enter vector A:
1 2 3
Enter vector B:
4 5 6
Addition done by Thread-26
Subtraction done by Thread-20
Multiplication done by Thread-3

Addition:
5 7 9

Subtraction:
-3 -3 -3

Multiplication:
4 10 18

Serial Time    = 0.000000756 seconds
Parallel Time  = 0.011006176 seconds
Speedup        = 0.0001
Efficiency     = 0.00%
```